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Nitte (Deemed to be University)

(Established under Sec (3) of UGC ACT,1956)

First Semester MCA (CBCS) Degree Examinations, Dec. 2025 - Jan. 2026**25MCA105 – RESEARCH METHODOLOGY & PUBLICATION ETHICS**

Duration: 3 Hours]

[Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Unit**.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Explain the meaning, objectives and characteristics of research with suitable examples.	10	L2	1	1
b)	Differentiate between Research Methods and Research Methodology with examples.	10	L2	1	1
2. a)	Describe the steps involved in defining a research problem and explain why defining the problem is crucial in research.	10	L2	1	1
b)	Distinguish between different types of research: Descriptive vs. Analytical, Applied vs. Fundamental.	10	L2	1	1
Unit – II					
3. a)	Discuss how digital libraries and AI-based tools assist in conducting a literature review.	10	L2	2	2
b)	Develop a research plan for analyzing the adoption of AI-based assistants in research.	10	L3	2	2
4. a)	Describe the steps involved in defining a research problem and explain why defining the problem is crucial in research.	10	L2	2	2
b)	Identify the role of literature review in developing a research problem.	10	L3	2	2
Unit – III					
5. a)	Outline the different sampling techniques used in research and their merits and demerits.	10	L2	3	2
b)	You are conducting a survey on the usage of e-learning platforms among postgraduate students. Design an appropriate research methodology and sampling technique for this study.	10	L3	3	2
6. a)	Illustrate the process of data collection using different instruments with examples.	10	L2	3	2
b)	A company wants to study employee satisfaction levels post-implementation of hybrid work. Design a suitable research design and sampling method.	10	L3	3	2
Unit – IV					
7. a)	Discuss the importance of ethics in scientific research and describe plagiarism detection tools used in research writing.	10	L2	4	2
b)	You are asked to mentor students on research ethics. Design a short code of conduct for them.	10	L3	4	2
8. a)	Summarize the types and consequences of plagiarism in research.	10	L2	4	2



- b) A researcher unintentionally reuses data from a previous publication without citation. Identify the ethical issues involved and suggest corrective measures.

10 L3 4 2

Unit – V

9. a) Explain various types of Intellectual Property Rights (IPR) relevant to research publications.
b) Identify the differences between open access and paid journals. Also describe the process of identifying predatory journals and their risks to researchers.
10. a) Write a note on TRIPS agreement.
b) You have developed a novel algorithm for detecting plagiarism in academic writing. Discuss the steps for protecting this innovation under IPR laws and filing a patent.

10 L2 5 3

10 L3 5 3

10 L2 5 3

10 L3 5 3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
